PAPER_685: UQFF Implications for Primordial Black Holes as Dark Matter

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Class: `UQFFPBHDarkMatterImplications` | CP4 #269
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Domain: Dark Matter / PBH Cosmology

Abstract

The UQFF extended PBH lifetime shifts the critical survival mass from $M_{\text{crit,std}} \approx 5 \times 10^{11}$ kg to $M_{\text{crit,UQFF}} = M_{\text{crit,std}}/\tau_{\text{ratio}}^{1/3} \approx 0.46 M_{\text{crit,std}}$, expanding the viable PBH dark matter mass window. The DM fraction is boosted: $f_{\text{PBH,UQFF}} = f_{\text{PBH,GR}} \cdot \tau_{\text{ratio}}^{2/3}$. In the UQFF framework the mass window 10^{10}–10^{17} kg is fully open as dark matter.

Primary UQFF Equation

$$M_{\text{crit,UQFF}} = \frac{M_{\text{crit,std}}}{\tau_{\text{ratio}}^{1/3}}, \quad f_{\text{PBH,UQFF}} = f_{\text{PBH,GR}} \cdot \tau_{\text{ratio}}^{2/3}$$

Parameters

Parameter	Value	Description
$M_{\text{crit,std}}$	5×10^{11} kg	Standard survival mass
τ_{ratio}	~ 11	UQFF lifetime boost
DM window	10^{10}–10^{17} kg	UQFF viable range

UQFF Constants

Constant	Value	Description
ρ_{UA}	7.09×10^{-36} kg/m³	Universal Aether density
ρ_{SCm}	7.09×10^{-37} kg/m³	Schwarzschild condensate

Constant	Value	Description
f_{TRZ}	0.1	Time-reversal zone factor
κ	5×10^{-4} day⁻¹	UQFF calibration constant
$[\mathrm{SSq}]$	0.57	Superstring quenching factor
μ_J	3.38×10^{23} J·m	Magnetic string coupling
γ	5×10^{-5} / 86400 s⁻¹	Aether oscillation decay

C++ Implementation

```cpp
#include "UQFFPBHDarkMatterImplications.h"

UQFFPBHDarkMatterImplications obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```python
from CondensedPhysics2 import UQFFPBHDarkMatterImplicationsCalculator

calc = UQFFPBHDarkMatterImplicationsCalculator()
result = calc.compute()  # returns all UQFF quantities
print(result)
```

References

Carr & Hawking 1974; Carr et al. 2021 (PBH DM); Raidal et al. 2019; UQFF Session 173

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